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```
> #Date: November 19, 2024
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```
> #This code reads in the Car Mileage Data Set. It then creates a  
frequency
```

```
> #histogram for miles per gallon by country and then a frequency  
table for miles
```

```
> #per gallon. It creates a stem and leaf plot for miles per gallon  
values, and then
```

```
> #finds descriptive statistics of miles per gallon by country through  
a new data frame.
```

```
> #Finally, probabilities are calculated using the standard normal  
distribution.
```

```
>
```

```
> #Reading in Car Mileage data set
```

```
> mpg<-read.csv("Car Mileage.csv")
```

```
> mpg
```

```
Country MPG
```

```
1 United States 18
```

```
2 United States 15
```

```
3 United States 18
```

```
4 United States 16
```

```
5 United States 17
```

```
6 United States 15
```

```
7 United States 14
```

```
8 United States 14
```

```
9 United States 14
```

```
10 United States 15
```

```
11 United States 15
```

```
12 United States 14
```

```
13 United States 15
```

14 United States 14
15 United States 22
16 United States 18
17 United States 21
18 United States 21
19 United States 10
20 United States 10
21 United States 11
22 United States 9
23 United States 28
24 United States 25
25 United States 19
26 United States 16
27 United States 17
28 United States 19
29 Japan 24
30 Japan 27
31 Japan 27
32 Japan 25
33 Japan 31
34 Japan 35
35 Japan 24
36 Japan 19
37 Japan 28
38 Japan 23
39 Japan 27
40 Japan 20
41 Japan 22

```
42   Japan 18
43   Japan 20
44   Japan 31
45   Japan 32
46   Japan 31
47   Japan 32
48   Japan 24
49   Japan 26
50   Japan 29
51   Japan 24
52   Japan 24
53   Japan 33
54   Japan 33
55   Japan 32
56   Japan 28
```

```
>
```

```
> #Plotting a frequency histogram for miles per gallon by country
```

```
> ggplot(mpg, aes(x=MPG))+
```

```
+ geom_histogram(color="yellow", fill="blue", binwidth=3)+
```

```
+ facet_wrap(vars(Country))
```

```
>
```

```
> #Producing a frequency table for miles per gallon
```

```
> mileage_freq_table<-hist(mpg$MPG, plot=F, breaks=6)
```

```
> mileage_freq_matrix<-matrix(c(mileage_freq_table$mids,
mileage_freq_table$counts), ncol=2)
```

```
> mileage_freq_matrix
```

```
  [,1] [,2]
```

```
[1,] 7.5  3
```

```
[2,] 12.5 11
```

```
[3,] 17.5 13
```

```
[4,] 22.5 12
```

```
[5,] 27.5 8
```

```
[6,] 32.5 9
```

```
> mileage_tab_frame<-data.frame(mileage_freq_matrix)
```

```
> colnames(mileage_tab_frame)<-c("Mileage Class Midpoints",  
"Frequency")
```

```
> mileage_tab_frame
```

	Mileage Class Midpoints	Frequency
--	-------------------------	-----------

1	7.5	3
---	-----	---

2	12.5	11
---	------	----

3	17.5	13
---	------	----

4	22.5	12
---	------	----

5	27.5	8
---	------	---

6	32.5	9
---	------	---

```
>
```

```
> #Stem and leaf diagram for the miles per gallon values
```

```
> stem(mpg$MPG)
```

The decimal point is 1 digit(s) to the right of the |

```
0 | 9
```

```
1 | 00144444
```

```
1 | 5555566778888999
```

```
2 | 001122344444
```

```
2 | 55677788889
```

```
3 | 11122233
```

```
>
```

```
> #Data frame with two columns: one with the MPG for United States  
and
```

```
> #the other with the MPG for Japan
```

```
> mpg_by_country<-data.frame(mpg$MPG[mpg$Country=="United  
States"], mpg$MPG[mpg$Country=="Japan"])
```

```
> colnames(mpg_by_country)<-c("UnitedStates", "Japan")
```

```
> mpg_by_country
```

	UnitedStates	Japan
1	18	24
2	15	27
3	18	27
4	16	25
5	17	31
6	15	35
7	14	24
8	14	19
9	14	28
10	15	23
11	15	27
12	14	20
13	15	22
14	14	18
15	22	20
16	18	31
17	21	32
18	21	31

```
19      10  32
20      10  24
21      11  26
22       9  29
23      28  24
24      25  24
25      19  33
26      16  33
27      17  32
28      19  28
```

```
>
```

```
> #Descriptive statistics of MPG for each country
```

```
> summary_stats_mpg<-summary(mpg_by_country)
```

```
> summary_stats_mpg
```

```
UnitedStates  Japan
```

```
Min. :9.00 Min. :18.00
```

```
1st Qu.:14.00 1st Qu.:24.00
```

```
Median :15.50 Median :27.00
```

```
Mean :16.43 Mean :26.75
```

```
3rd Qu.:18.25 3rd Qu.:31.00
```

```
Max. :28.00 Max. :35.00
```

```
>
```

```
> #Finding P(Z<0), with graph
```

```
> prob_lt0<-pnormGC(0, region="below", graph=TRUE)
```

```
> prob_lt0
```

```
[1] 0.5
```

```
>
```

```
> #Finding P(Z>0), with graph
```

```
> prob_gt0<-pnormGC(0, region="above", graph=TRUE)
```

```
> prob_gt0
```

```
[1] 0.5
```

```
>
```

```
> #Finding  $P(-1.96 < Z < 1.96)$  to 2 decimal places, with graph
```

```
> prob_btw1.96<-round(pnormGC(c(-1.96, 1.96), region="between",  
graph=TRUE), 2)
```

```
> prob_btw1.96
```

```
[1] 0.95
```

```
>
```